Clustering Indian States and Union Territories Using SDG India Index Goal Scores

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Abstract

The Sustainable Development Goals (SDGs) provide a framework for evaluating progress in areas such as health, education, equality, environmental sustainability, and economic development. This project applies unsupervised machine learning techniques to explore similarities among Indian States and Union Territories using their SDG India Index 2023–24 goal scores. The dataset contains 36 regions and scores for 16 SDG goals. Goal 14 was excluded from the clustering analysis because 27 of its 36 values were missing, leaving 15 goal-score features for analysis. The selected features were standardised before applying K-Means and hierarchical clustering for cluster counts ranging from two to eight. The models were evaluated using the Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Score. Principal Component Analysis (PCA) was used to visualise the clustered regions, while correlation analysis examined relationships among SDG goal scores. K-Means achieved its highest Silhouette Score of 0.1967 at K = 2, while hierarchical clustering achieved its highest score of 0.1803 at K = 3. The first two principal components explained approximately 42.47% of the variance. The results reveal exploratory patterns in regional SDG performance, although the relatively low Silhouette Scores indicate that cluster separation is limited.

Keywords: Sustainable Development Goals, SDG India Index, Machine Learning, K-Means, Hierarchical Clustering, PCA, Correlation Analysis.

1. Introduction

The Sustainable Development Goals (SDGs) were established to address major global challenges, including poverty, hunger, health, education, gender equality, clean energy, sustainable cities, and environmental protection. India monitors progress towards these goals through the SDG India Index published by NITI Aayog.

Different States and Union Territories may show different levels of performance across individual SDGs. A region may perform well in health and education but face challenges in employment, equality, or environmental sustainability. Examining each goal separately provides useful information, but it may not clearly reveal broader similarities between regions.

Machine learning can help explore these similarities by grouping regions according to their goal-score patterns. This project uses unsupervised learning because the regions are not provided with predefined cluster labels. K-Means and hierarchical clustering are applied to identify groups with similar SDG performance profiles.

The project also uses Principal Component Analysis (PCA) to represent high-dimensional data in two dimensions and correlation analysis to examine relationships among goal scores. The purpose is exploratory: the project aims to understand patterns in the available data rather than predict future SDG scores or establish causal relationships.

1.1 Objectives

The main objectives of this project are:

To prepare and analyse the SDG India Index 2023–24 data for Indian States and Union Territories.

To apply K-Means and hierarchical clustering to group regions with similar SDG goal-score patterns.

To compare clustering results using standard internal evaluation metrics.

To visualise regional clusters using PCA.

To examine correlations among SDG goals and interpret the findings.

To identify limitations and possible improvements for future analysis.

2. Dataset and Preprocessing

2.1 Dataset Description

The dataset was prepared using the State-wise and Union Territory-wise SDG India Index 2023–24 score data. It contains 36 regions, including States and Union Territories, with scores for 16 SDG goals and an overall composite score.

Each row represents a region, and each goal column represents its score for a particular SDG. The goal scores are used as the main features for clustering. The composite score is retained as a descriptive field and is not included among the 15 goal-score features used in the clustering analysis.

Table 1. Summary of the dataset

Attribute	Description
Data source	SDG India Index 2023–24
Number of regions	36
Original goal-score features	16
Goal excluded	Goal 14
Missing values in Goal 14	27 out of 36
Goal-score features used	15
Main analysis	Clustering and correlation
Visualisation method	PCA

2.2 Data Preprocessing

Data preprocessing was performed before applying the clustering algorithms. The State-wise and Union Territory-wise records were prepared in a common dataset. Region names and region types were retained to make the results interpretable.

Goal 14 was excluded from the clustering features because 27 of its 36 values were missing. Including a feature with such extensive missingness could make the analysis less reliable. The remaining 15 goal-score features were used for clustering.

The selected features were standardised using feature scaling. Standardisation places features on a comparable scale, helping prevent differences in their numerical ranges from disproportionately influencing distance-based clustering methods.

The resulting dataset was used for K-Means clustering, hierarchical clustering, PCA visualisation, and correlation analysis. Missing values and preprocessing decisions should be interpreted in the context of the available source data.

3. Methodology

The project follows a sequence of data preparation, feature standardisation, clustering, evaluation, visualisation, and interpretation.

3.1 K-Means Clustering

K-Means is an unsupervised machine learning algorithm that divides observations into a specified number of clusters. It assigns each region to the cluster whose centroid is closest to it, based on the selected feature space.

The algorithm repeatedly updates cluster assignments and centroids until the assignments stabilise or a stopping condition is reached. In this project, K-Means was evaluated for cluster counts from K = 2 to K = 8. The resulting groups represent regions with similar patterns across the selected SDG goal scores.

3.2 Hierarchical Clustering

Hierarchical clustering builds a hierarchy of clusters based on similarities or distances between observations. The hierarchy can be represented using a dendrogram, which illustrates how individual regions or smaller clusters combine into larger groups.

In this project, hierarchical clustering was evaluated for K values from 2 to 8. Its results were compared with those of K-Means to examine whether the two methods produced similar clustering quality.

3.3 Principal Component Analysis

Principal Component Analysis (PCA) is a dimensionality-reduction technique that transforms a set of potentially correlated features into a smaller number of components. These components retain as much variation in the data as possible, in descending order.

PCA was used to visualise the regions in two dimensions. The first principal component explained 27.27% of the variance, and the second explained 15.20%. Together, the first two components explained approximately 42.47% of the total variance. Therefore, the two-dimensional plot provides a useful visual summary but does not represent all the information contained in the original features.

3.4 Correlation Analysis

Correlation analysis was used to examine the strength and direction of relationships between SDG goal scores. Positive correlation indicates that two goal scores tend to increase together, while negative correlation indicates that higher values in one tend to be associated with lower values in the other.

A correlation heatmap was generated to make these relationships easier to examine. Correlation does not establish causation, and the observed relationships should not be interpreted as proof that changes in one SDG directly cause changes in another.

3.5 Evaluation Metrics

The clustering models were compared using three internal evaluation metrics:

- Silhouette Score: Measures how similar a region is to its own cluster compared with other clusters. Higher values generally indicate better separation.
- Davies–Bouldin Index: Measures similarity between clusters in terms of their spread and separation. Lower values generally indicate better clustering.
- Calinski–Harabasz Score: Compares between-cluster dispersion with within-cluster dispersion. Higher values generally indicate more distinct clusters.

No single metric fully describes clustering quality. Therefore, all three metrics were considered during model comparison.

4. Experimental Results

K-Means and hierarchical clustering were tested using cluster counts ranging from two to eight. Their Silhouette Scores, Davies–Bouldin Index values, and Calinski–Harabasz Scores are below :

Table 2. K-Means clustering evaluation

K	Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Score
2	0.1967	1.7964	9.5119
3	0.1850	1.7595	7.3515
4	0.1545	1.8197	6.6071
5	0.1473	1.6092	6.1412
6	0.1571	1.3054	5.7591
7	0.1789	1.2192	5.8979
8	0.1673	1.2842	5.6008

Table 3. Hierarchical clustering evaluation

K	Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Score
2	0.1737	2.0057	7.8343
3	0.1803	1.6091	6.9233
4	0.1324	1.8050	6.2950
5	0.1561	1.5960	6.1192
6	0.1736	1.3856	6.1466
7	0.1697	1.3639	5.8215
8	0.1680	1.2311	5.6443

4.1 Comparison of Clustering Methods

K-Means achieved its highest Silhouette Score of 0.1967 at K = 2. Its highest Calinski–Harabasz Score, 9.5119, was also obtained at K = 2. However, its lowest Davies–Bouldin Index of 1.2192 occurred at K = 7.

Hierarchical clustering achieved its highest Silhouette Score of 0.1803 at K = 3. Its highest Calinski–Harabasz Score of 7.8343 occurred at K = 2, while its lowest Davies–Bouldin Index of 1.2311 occurred at K = 8.

These results show that the preferred cluster count depends partly on the evaluation metric. K-Means performs better than hierarchical clustering on the highest Silhouette Score observed in these experiments. However, neither method produces strongly separated clusters according to the Silhouette Scores. The findings should therefore be treated as exploratory rather than as a definitive classification of regional development.

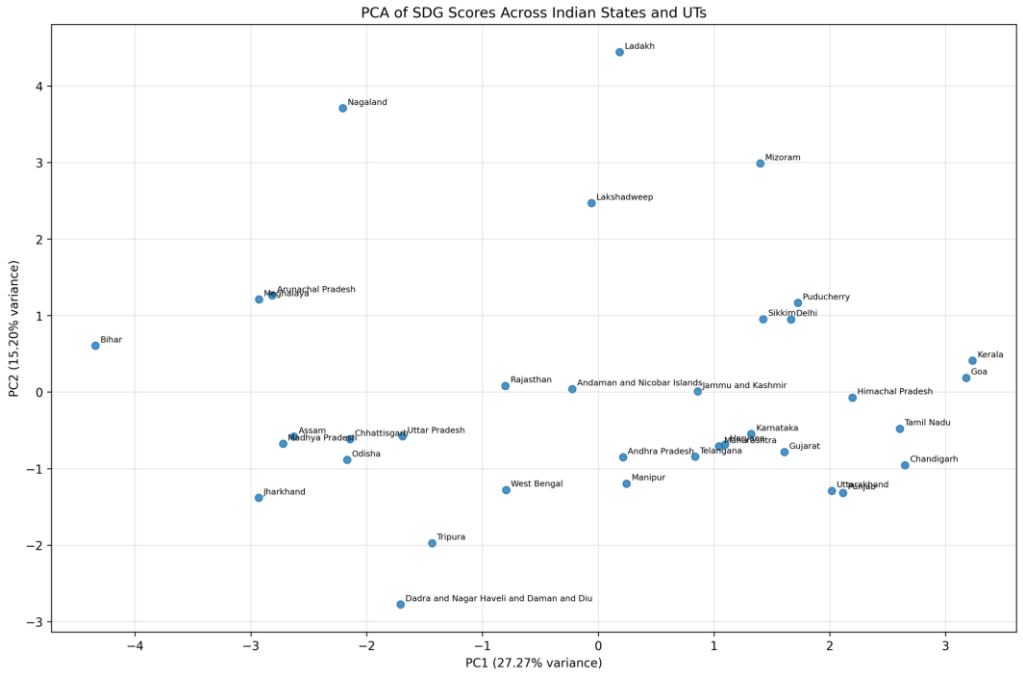


Figure 1. PCA visualisation of regions according to their SDG goal-score patterns and cluster assignments.

The PCA visualisation provides a two-dimensional view of the regional groupings. Because the first two principal components explain approximately 42.47% of the variance, some differences between regions may not be visible in this plot.

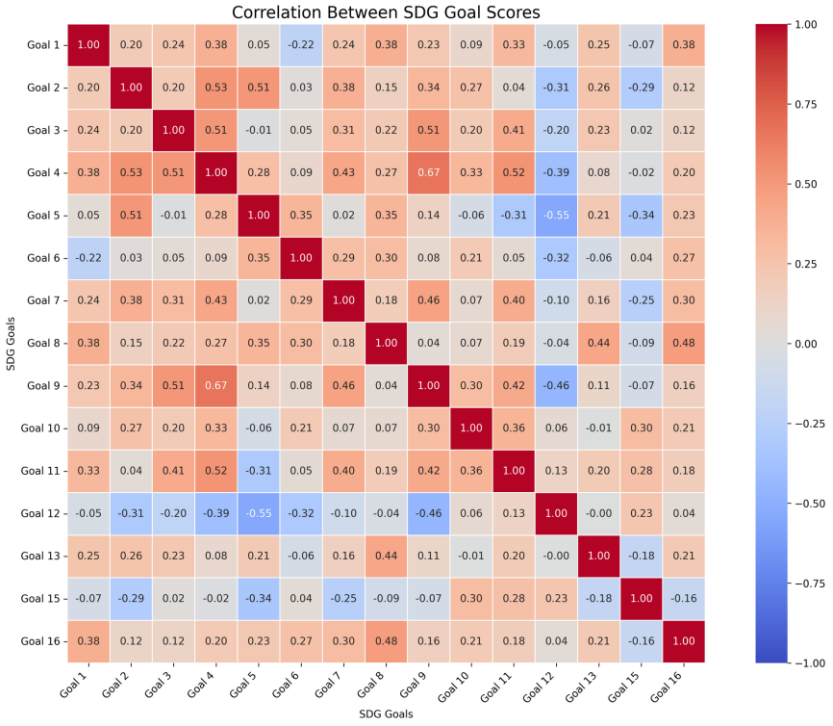


Figure 2. Correlation heatmap showing relationships among the SDG goal scores.

The heatmap helps identify goal pairs with relatively strong positive or negative correlations. These patterns describe associations in the dataset and should not be interpreted as causal relationships.

5. Discussion

The experimental results indicate that clustering can be used to explore similarities among Indian States and Union Territories based on their SDG goal scores. K-Means produced its highest Silhouette Score at K = 2, while hierarchical clustering produced its highest Silhouette Score at K = 3. However, the relatively low scores suggest that the regions do not form clearly separated groups in the selected feature space.

For further interpretation, the K-Means solution with K = 2 was examined. Cluster 0 contained 14 regions, while Cluster 1 contained 22 regions. The average scores for most selected goals were higher in Cluster 1. For example, the average Goal 2 score was 48.57 in Cluster 0 and 69.18 in Cluster 1. The average Goal 4 score was 51.43 in Cluster 0 and 68.95 in Cluster 1. The average Goal 9 score was 47.43 in Cluster 0 and 61.00 in Cluster 1.

There were exceptions to this pattern. Cluster 0 had higher average scores for Goal 12 and Goal 15. These differences indicate that regional performance varies across individual SDGs and that a single cluster should not be treated as a complete measure of overall development.

The correlation analysis also revealed relationships between goal scores. The strongest positive correlation observed was between Goals 4 and 9, at approximately +0.67. A relatively strong negative correlation was observed between Goals 5 and 12, at approximately −0.55. Other positive relationships included Goals 2 and 4, at +0.53; Goals 4 and 11, at +0.52; and Goals 3 and 9 and Goals 2 and 5, at approximately +0.51 each.

These relationships may help identify patterns for further investigation, but correlation alone cannot explain why the patterns occur. Additional socioeconomic, geographic, and policy-related information would be needed to investigate possible explanations.

6. Limitations and Future Scope

This project has several limitations. First, the analysis includes only 36 States and Union Territories, so the number of observations is relatively small for clustering. Second, Goal 14 was excluded because 27 of its 36 values were missing. This means the analysis does not represent all 16 SDG goals.

Third, the Silhouette Scores were relatively low, suggesting that the clusters overlap and are not strongly separated. Different clustering algorithms, distance measures, and cluster counts may produce different groupings. Fourth, the first two PCA components explain only 42.47% of the variance, so the two-dimensional visualisation cannot represent every difference among regions.

Future work could compare additional clustering algorithms, investigate suitable methods for handling missing data, and analyse SDG scores across multiple years to examine changes over time. Additional socioeconomic indicators could also be considered to provide broader context. Any such extensions would require further data preparation and evaluation before conclusions could be drawn.

7. Conclusion

This project applied K-Means and hierarchical clustering to explore similarities among 36 Indian States and Union Territories using SDG India Index 2023–24 goal scores. After excluding Goal 14 because of extensive missing values, 15 goal-score features were standardised and used for analysis. The models were evaluated for cluster counts from two to eight using the Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Score.

K-Means achieved the highest Silhouette Score of 0.1967 at K = 2, while hierarchical clustering achieved 0.1803 at K = 3. PCA was used to visualise the regional patterns, and correlation analysis examined relationships among SDG goals. The results show that clustering can support exploratory analysis of regional SDG performance, but the relatively low Silhouette Scores require cautious interpretation.

Overall, the project demonstrates how unsupervised machine learning can help organise and explore multidimensional development data. The resulting clusters are analytical groupings, not official development categories or definitive rankings of States and Union Territories.

Declarations

Data Availability: The analysis uses SDG India Index 2023–24 State-wise and Union Territory-wise score data published by NITI Aayog.

Conflict of Interest: The author declares no known conflict of interest.

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